

CH 1 — Graphs of Scalar Functions

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1.0 Parametric Curves

We have seen functions that are either explicit or implicit

1. $y = f(x)$
2. $x = y^2$
3. $x^2 + y^2 = 1$

We can also describe a curve using parametric equations where

$$x = x(t), y = y(t), t \in [a, b], \forall a, b \in \mathbb{Z}$$

with $(x(t), y(t))$ is a point of the curve and t is the parameter.

- The curve is called **parametric curve**.
- We get a direction in which the curve is traced as well.

Practices:

1. Find the parametric curve: $x = t - 1, y = 2t + 1, t \in [0, 3]$

$$x + 1 = t \implies y = 2(x + 1) + 1 = 2x + 3, x \in [-1, 2]$$

2. Find the parametric curve: $x = \cos(t), y = \sin(t), t \in [0, 2\pi]$

$$x^2(t) + y^2(t) = \cos^2 t + \sin^2 t = 1, t \in [0, 2\pi]$$

3. Find the parametric curve: $x = \cos(2t), y = \sin(2t), t \in [0, 2\pi]$

$$x^2 + y^2 = \cos^2(2t) + \sin^2(2t) = 1, t \in [0, 2\pi]$$

Same curve as Q2 but has two different parametrizations. Twice as the speed as the Q2.

1.1 Vector-Valued Functions

Definition — Vector-Valued Function

It is a function whose domain is a subset of the $\mathbb{R}$ and whose output is a vector

Note: a vector-valued function is a convenient way to collect scalar functions

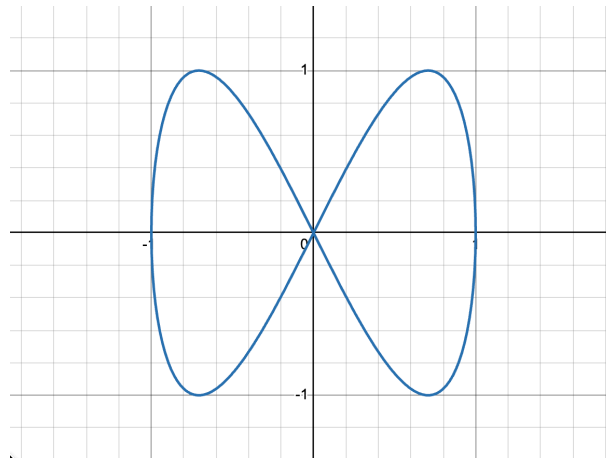
Examples:

1. In $\mathbb{R}^2$, $\vec{x}(t) = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$

2. In $\mathbb{R}^3$, $\vec{x}(t) = \begin{bmatrix} x(t) \\ y(t) \\ z(t) \end{bmatrix}$

3. Let $\vec{x}(t) = (\cos t, \sin 2t)$ represent the position of a particle at time t . Sketch $\vec{x}(t), 0 \leq t \leq 2\pi$

We have $x = \cos t, y = \sin 2t, 0 \leq t \leq 2\pi$



Definition – Velocity

$$\text{Velocity} = \vec{v}(t) = \frac{d\vec{x}(t)}{dt}$$

Examples:

1. $\vec{x}(t) = (\cos t, \sin 2t)$

$$\text{velocity} = \frac{d\vec{x}(t)}{dt} = (-\sin t, 2 \cos 2t)$$

$$\text{At } t = 0, \vec{v}(0) = (0, 2) \quad \text{At } t = \frac{\pi}{2}, \vec{v}\left(\frac{\pi}{2}\right) = (-1, 0)$$

2. Let $\vec{x}_1(t) = (\cos t, \sin t)$ and $\vec{x}_2(t) = (\cos 3t, \sin 3t)$ be the paths of two particles, find the velocity of each particle when $t = 0$

$$\frac{d\vec{x}_1(t)}{dt} = (-\sin t, \cos t), \text{ 1 revolution of unit circle radius 1 unit}$$

$$\frac{d\vec{x}_2(t)}{dt} = (-3 \sin 3t, 3 \cos 3t), \text{ 3 revolutions of unit circle radius 1 unit}$$

$$\vec{x}_1'(0) = (0, 1), \vec{x}_2'(0) = (0, 3)$$

Definition – Tangent vector

Given the curve defined by

$$\vec{x}(t) = (x(t), y(t))$$

its **tangent vector** at $t = t_0$ is

$$\vec{t} = \left. \frac{d\vec{x}}{dt} \right|_{t=t_0} = (x'(t_0), y'(t_0))$$

whenever $\frac{d\vec{x}}{dt}, t = t_0 \neq \vec{0}$ exists.

Note: $\vec{t}$ lies on the tangent line and points in the direction of travel

Example:

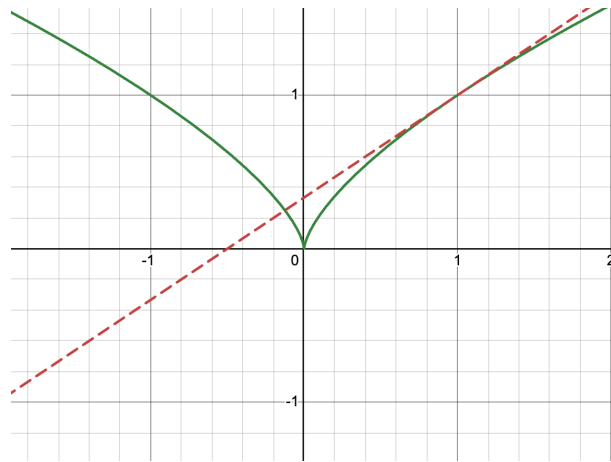
Consider the curve $\vec{x}(t) = (t^3, t^2)$ and its tangent vector.

Notice that if $x = t^3 \rightarrow t = x^{\frac{1}{3}} \rightarrow y = x^{\frac{2}{3}}$

$$\frac{d\vec{x}(t)}{dt} = (3t^2, 2t)$$

$$t = 1, \vec{x}(1) = (1, 1), \vec{v}(1) = (3, 2)$$

$t = 0, \vec{x}(0) = (0, 0), \vec{v}(0) = (0, 0) = \vec{0}$ We cannot define the tangent vector as the function at the point is not differentiable



1.2 Scalar Functions

Recall:

$f : A \rightarrow B$ is a function

1. Each element $a \in A$ is a unique element $f(a) \in B$ called the **image** of a under f
2. The set A is called the **domain** of f and is denoted by $D(f)$
3. The set B is called the **codomain** of f .
4. The subset of B consisting of all $f(a)$ is called the **range** of f , denoted by $R(f)$
5. Single variable scalar functions are $f : \mathbb{R} \rightarrow \mathbb{R}$

Definition – Scalar Function

A **scalar function** $f(x_1, \dots, x_n)$ of n -variables is a function whose domain is a subset of $\mathbb{R}^n$ and whose range is a subset of $\mathbb{R}$

Examples:

1. $f(x, y) = x^2 + y^2 \rightarrow f(1, 1) = 1^2 + 1^2 = 2$
2. $f(x, y) = \sin(x + y) \rightarrow f(0, \frac{\pi}{2}) = \sin(\frac{\pi}{2}) = 1$
3. $f(x, y, z) = e^{x+y+z} \rightarrow f(1, 2, 1) = e^{1+2+1} = e^4$

Notation: We can write $\vec{x} = (x_1, x_2, \dots, x_n)$ so that $f(\vec{x}) = f(x_1, x_2, \dots, x_n)$

Our independent variables don't have to be x, y, z , etc... Multivariable functions can be used to model many different physical situations

Example: write down a multivariable function which calculates the volume of a cylinder.

$$V = \pi r^2 h \rightarrow V(r, h) = \pi r^2 h$$

Definition – Domain and Range

Domain is the largest subset of $\mathbb{R}^n$ for which the function is defined

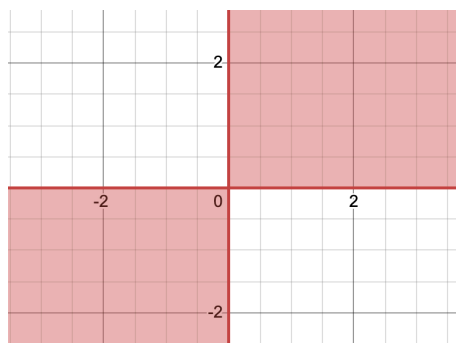
Range is a set of all possible values which are subset of $\mathbb{R}$

Examples:

1. Find the domain and range of $f(x, y) = \sqrt{xy}$

$$\text{We need } xy \geq 0 \Rightarrow \begin{cases} x \geq 0 \text{ and } y \geq 0 \\ x \leq 0 \text{ and } y \leq 0 \end{cases}$$

$$\text{dom}(f) = \{(x, y) \in \mathbb{R}^2 : x \geq 0, y \geq 0\} \cup \{(x, y) \in \mathbb{R}^2 : x \leq 0, y \leq 0\}$$



We know that $\sqrt{xy} \geq 0, \forall x, y \in \text{dom}(f)$

Let $c \in \mathbb{R}, c \geq 0$, be an arbitrary non-negative real number.

Notice that $f(c^2, 1) = \sqrt{c^2} = |c| = c$ since $c \geq 0$

For any $c \in \mathbb{R}, c \geq 0, \exists$ inputs in the domain of f s.t. c is a valid output $\rightarrow c \in \text{ran}(f) : \mathbb{R}_{\geq 0}$

2. Find the domain and range of $g(x, y) = \frac{x^2 - y^2}{|x| + |y|}$

We need $|x| + |y| \neq 0$ which is true $\forall x, y \in \mathbb{R}$ both not zero since $|x| > 0, |y| > 0$

$$\text{dom}(g) = \{(x, y) : x, y \in \mathbb{R} \setminus \{(0, 0)\}\}$$

Consider case $x > y: x^2 - y^2 > 0, |x| + |y| > 0 \Rightarrow g(x, y) > 0$

Consider case $x = y: x^2 - y^2 = 0, |x| + |y| = 2x \Rightarrow g(x, y) = 0$

Consider case $x < y: x^2 - y^2 < 0, |x| + |y| > 0 \Rightarrow g(x, y) < 0$

or alternatively

$$\text{Let } c \in \mathbb{R}, c > 0, g(c, 0) = \frac{c^2 - 0}{|c| + 0} = |c| = c, c > 0$$

$$\text{Let } d \in \mathbb{R}, d < 0, g(0, d) = \frac{0 - d^2}{0 + |d|} = -|d| = d, d < 0$$

$$\text{Let } a, b \in \mathbb{R}, a = b, g(a, b) = \frac{a^2 - b^2}{|a| + |b|} = \frac{a^2 - a^2}{|a| + |a|} = \frac{0}{2a} = 0$$

Thus $\text{ran}(g) = \mathbb{R}$

1.3 Geometric Interpretation of $z = f(x, y)$

Question: How we can graph $z = f(x, y)$

ANS: We graph $(a, b, f(a, b))$. We think of $f(a, b)$ as height above the xy -plane

Definition – Levels

1. The **level curves** of a function $f(x, y)$ are the curves

$$f(x, y) = k$$

where k is a constant in the range of f

2. A **level surface** of a scalar function $f(x, y, z)$ is defined by

$$f(x, y, z) = k, \quad k \in R(f)$$

3. A **level set** of a scalar function $f(x), x \in \mathbb{R}^n$ is defined by

$$f(x) = k, \quad k \in R(f)$$

Definition – Cross Sections

A **cross section** of a surface $z = f(x, y)$ is the intersection of $z = f(x, y)$ with a plane